

OpenClaw Temperature and Humidity Dashboard Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the temperature and humidity dashboard project itself.

1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capability in the desktop dashboard scenario through the combination of "sensor + display + indication". After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, RGB, and servo into a temperature and humidity dashboard
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four project scenarios: real-time reading, OLED meter display, RGB status light, servo pointer mapping

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display screen
- RGB light strip / RGB LED module
- [optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Temperature and humidity reading command: `/home/pi/hardware_hal_demo/dht`
- OLED control command: `/home/pi/hardware_hal_demo/oled`
- RGB control command: `/home/pi/hardware_hal_demo/rgb`
- Servo control command: `/home/pi/hardware_hal_demo/cli/servo`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

This project is very similar to the AI weather station, but with different focus:

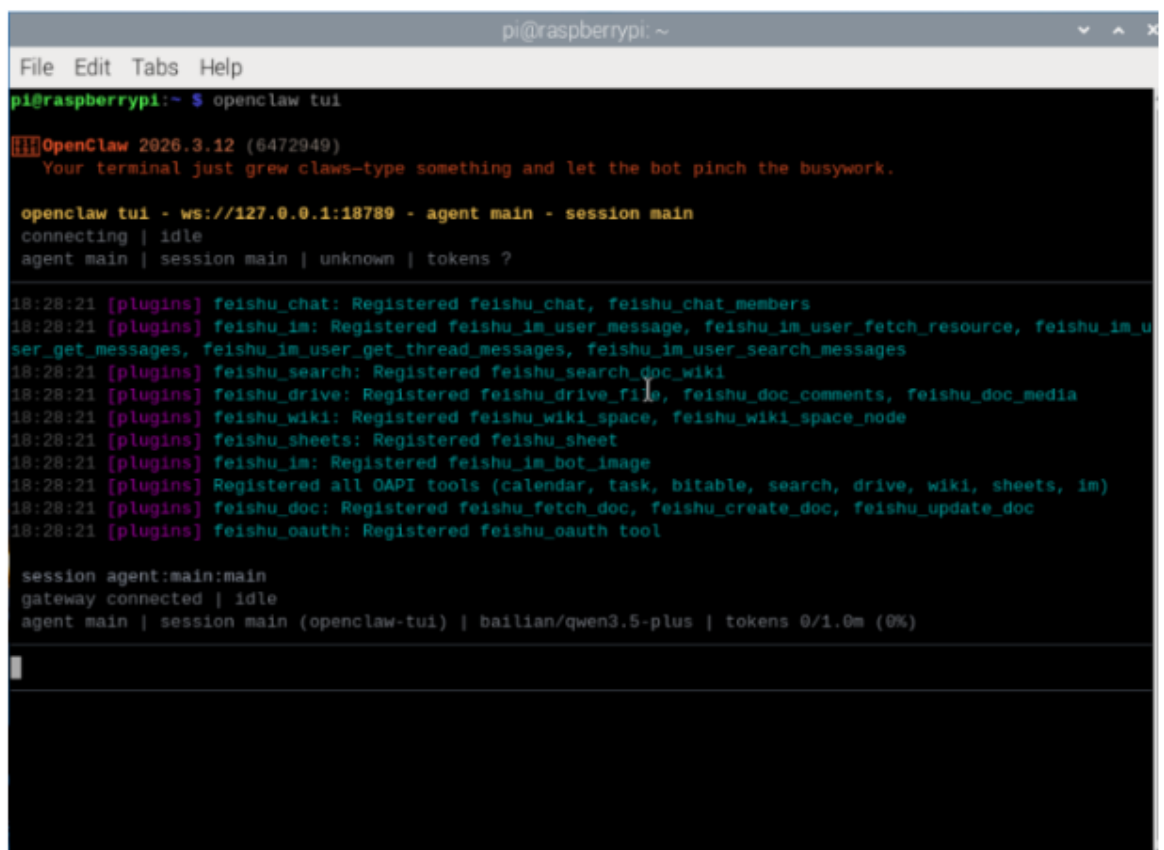
- AI weather station emphasizes "broadcast" and "explanation"
- Temperature and humidity dashboard emphasizes "display" and "status mapping"

4. Testing Project Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, proceed with the temperature and humidity dashboard-specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```



```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ openclaw tui  
[!!!] OpenClaw 2026.3.12 (6472949)  
Your terminal just grew claws-type something and let the bot pinch the busywork.  
  
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
connecting | idle  
agent main | session main | unknown | tokens ?  
  
18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members  
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u  
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages  
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki  
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media  
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node  
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet  
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image  
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)  
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc  
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool  
  
session agent:main:main  
gateway connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)  
  
|  
  
|
```

Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display current temperature and humidity on OLED
- If environment is normal, light up green light
- Make servo rotate to 90 degrees

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote status viewing, classroom demonstration	Convenient for remotely reading current panel data	Demonstrate "remote monitoring panel"
TUI	Local debugging, mapping verification	Easiest to troubleshoot reading and action relationships	First to get the visualization chain working
Voice Code	Voice broadcast, desktop assistant	More like daily usage experience	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Configuration Tests]. After creating, you can remotely check the dashboard status in WhatsApp, for example:

- Temperature and humidity dashboard

20:36



< 2



+86 153 3885 7526 (你)

给自己发消息

warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓

Temperature and humidity dashboard

20:35 ✓✓

Temperature and humidity dashboard is now active!



Current Readings:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 54%



OLED Display: Showing temp and humidity values

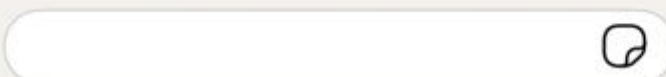


RGB Indicator: Red blinking light (speed 5)

- This indicates the temperature is **above 28°C** (warm/hot zone)

Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!

20:35 ✓✓



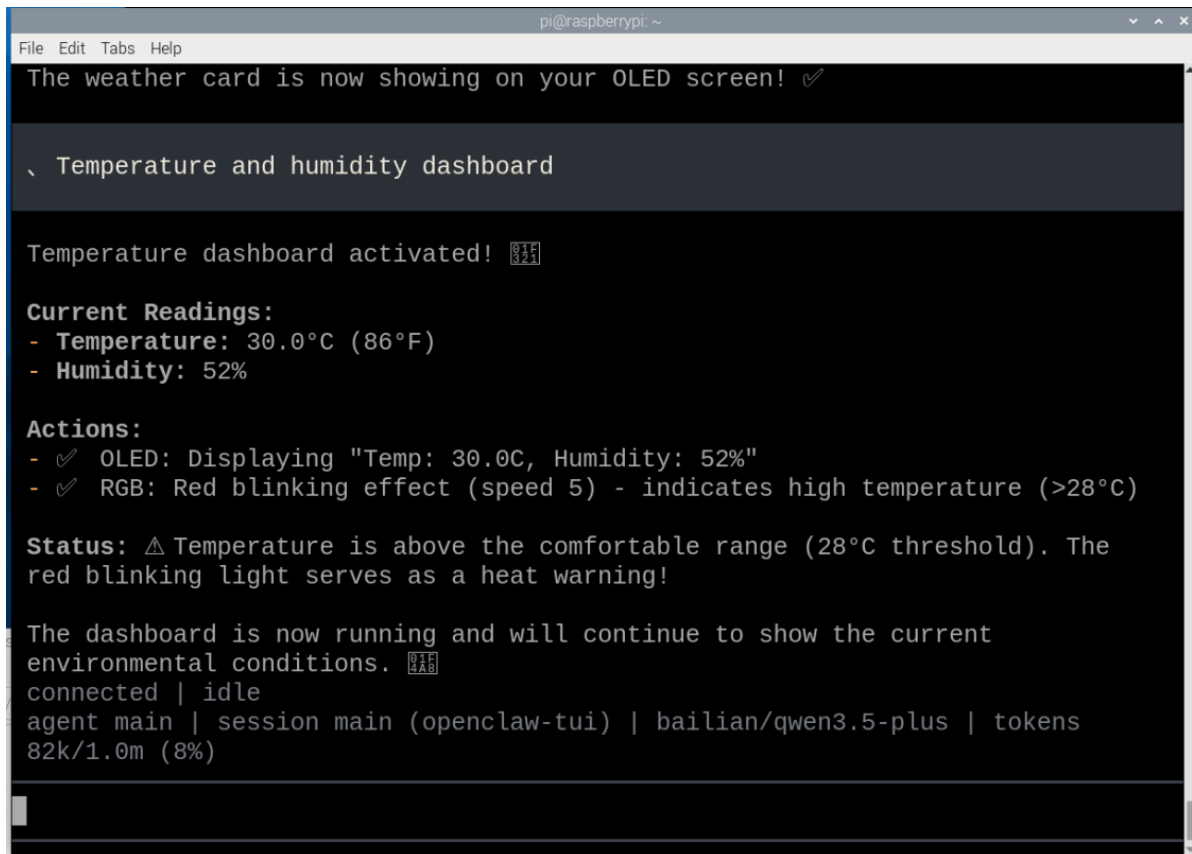
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then input project-related commands, for example:

- Temperature and humidity dashboard



```
pi@raspberrypi: ~  
File Edit Tabs Help  
The weather card is now showing on your OLED screen! ✓  
  
└─ Temperature and humidity dashboard  
  
Temperature dashboard activated! 🌡️  
  
Current Readings:  
- Temperature: 30.0°C (86°F)  
- Humidity: 52%  
  
Actions:  
- ✓ OLED: Displaying "Temp: 30.0C, Humidity: 52%"  
- ✓ RGB: Red blinking effect (speed 5) - indicates high temperature (>28°C)  
  
Status: ⚠️ Temperature is above the comfortable range (28°C threshold). The  
red blinking light serves as a heat warning!  
  
The dashboard is now running and will continue to show the current  
environmental conditions. 📺  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens  
82k/1.0m (8%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Temperature and humidity dashboard

```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Temperature and humidity dashboard.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 🌡️Temperature & Humidity Dashboard Active!

**Current
  Readings:**
- **Temperature:** 30.1°C (86.2°F)
- **Humidity:** 51%
  **Dashboard Status:**
- ✔️ OLED displaying real-time values
- ✔️ RGB
  indicator: **Red blinking** (heat warning - above 28°C)
  **Temperature Color Code:**
- 🌡️ Blue breathing: < 20°C (Cold)
- 🌡️
  Solid green: 20-28°C (Comfortable)
- 🌡️ Red blinking: > 28°C (Hot) ⚠️
  **Current Status:** **HOT** - Temperature is above the comfort
  threshold! Consider cooling measures. 🌡️

The dashboard will continue
  showing live readings. Want me to check again in a few minutes?
```

6. Project Implementation

Note: All demonstrations below use the `openclaw_tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Scenario 1: Real-time Reading

Experiment Objective

Read current temperature and humidity to complete the basic data input for the dashboard.

Effect Description

OpenClaw will first get temperature and humidity readings, which are the basis for subsequent OLED, RGB, and servo mapping.

Example Commands

- Read current temperature and humidity
- What are current temperature and humidity
- Help me refresh current readings

6.2 Scenario 2: OLED Meter Display

Experiment Objective

Display current temperature and humidity results as concise OLED dashboard content.

Effect Description

This is suitable for using `metrics` or `dht_oled` method to combine `Temp` and `Hum` into a compact screen area, forming a small data panel.

Example Commands

- Display Temp and Hum on OLED
- Make a simple temperature and humidity panel
- Write current temperature and humidity to the screen

6.3 Scenario 3: RGB Status Light

Experiment Objective

Express the current environment as normal, too hot, too cold, or too dry with RGB colors.

Effect Description

This phase can make the dashboard more intuitive. For example, light up green when normal, yellow when too hot, red when abnormal, blue when too cold - you can see the status at a glance.

Example Commands

- Light up green light if environment is normal
- Light up yellow light if temperature is high
- Light up red light if humidity is abnormal

6.4 Scenario 4: Servo Pointer Mapping

Experiment Objective

Map temperature or humidity to servo angles, making the servo point to a position like a dashboard pointer.

Effect Description

For example, map temperature range to 0 to 180 degrees. The higher the current temperature, the more the servo pointer moves to the right. This effect is very suitable for classroom demonstration because "numbers becoming physical actions" is very intuitive.

Example Commands

- Use servo to indicate current temperature like a pointer
- Map current temperature to servo angle
- Make the pointer point to current status